

Guangpu HUANG, Ph.D.

CONTACT INFORMATION	TF 401, ELEC Aalto University, Helsinki, Finland +358 (0)50 4388 962 gpuhuang@gmail.com Google Scholar
RESEARCH INTERESTS	• Multi-Modal Speech Recognition, Large Language Models • Semi-supervised Learning, Active Learning, Recurrent Neural Networks
SKILLS	• Python, Unix, Perl, Git, Jupyter, Google Cloud & Colabs • Tensorflow, Keras, PyTorch, Matplotlib, SciPy, NumPy, Pandas • ASR: HTK, KALDI, PRAAT, Wav2Vec • NLP: BERT, SRILM, RNNLM, Word2Vec, NLTK
EDUCATION	• Ph.D. Nanyang Technological University, Singapore Aug 2008 - Jul 2012 Electrical and Electronic Engineering Thesis: <i>Articulatory Phonetic Features for Improved Speech Recognition</i> • B.Eng. Nanyang Technological University, Singapore Aug 2004 - Jun 2008 First Class Honours in Electrical and Electronics Engineering Minor in English Literature Thesis: <i>Design and Development of a Software System to Assist Students in Speaking Standard English</i> • Linköping University of Technology, Sweden Aug 2005 - Jan 2006 Computer Science and Electronic Engineering
WORKING EXPERIENCES	• Finnish Center for Artificial Intelligence June 2023 - Present Postdoctoral Researcher – Lead research in multi-modal speech emotion recognition. Integrate research outputs from the respective field of ASR, NLP, and CV to achieve improved emotion recognition. * Propose research topics and supervise bachelor/master/PhD students on their work. * Publish research results in top-tier journals and conferences. – Coordinate efforts among research teams from Aalto University, University of Helsinki, and FCAI * Organize workshops * Deliver lectures • SILO.AI, Finland May 2020 - May 2023 Senior Scientist – Develop machine learning solutions for a mobile client with over 500 million users: * Implement state of the art multi-lingual Bert-based text classifiers to automatically detect toxic online comments. * Investigate the causes and effects of data sparsity and human disputes when annotating data. * Develop an active learning scheme to identify the positive toxic comments. On a Kaggle NLP challenge, it achieves the same or higher level test accuracy using a core-set of less than 10% of the original training data. – Conduct proof of concept studies on latest NLP and ASR research, and transfer speech technology to production. – Design, develop, and deploy speech recognition demos and software systems. – Submit patent applications on speech and language technologies. • Audio Research Group, Tampere University, Finland Dec 2016 - April 2020 Postdoctoral Researcher

- Collaborate with industrial partner on speech research and software development:
 - * Build a deep learning module to automatically cluster the pronunciation errors of English learners into meaningful categories.
 - * Develop an interactive machine learning API to assist linguists to accurately annotate audio corpus with representative audio samples.
 - * The method identifies the minimum number of data samples necessary to achieve the highest accuracy. It significantly reduces money and time cost when making massive datasets.
- Conduct research in environmental sound event detection using neural language models:
 - * Build neural networks to model the contextual dependancies in environmental sounds.
 - * Build language models to predict sound event labels.
- Supervise Master and Bachelor students on thesis works.

• **LIMSI-CNRS, Paris-Sud University, France**

Postdoctoral Researcher

Apr 2015 - Nov 2016

- Conduct research in low-resource language ASR using neural language models:
 - * Build a neural machine-translation model using parallel Mandarin-Cantonese TV subtitles to augment the language models with Cantonese text data.
 - * The method greatly reduces the language model perplexity on the low-resourced Cantonese language, less than 100 hours of training data. It also improved word recognition accuracy and keyword retrieval performance on 5 of the 7 languages in the BABEL project. Our work has been presented and published in 3 INTERSPEECH and ICASSP papers.

• **Temasek Laboratories, NTU, Singapore**

Research Scientist

Sep 2012 - Mar 2015

- Conduct research in robust ASR:
 - * Develop a preprocessing module to extract multi-view speech features: prosodic, acoustic, articulatory, and linguistic, to statistically model the sentence structure.
 - * Implement a segmentation module: deep neural networks with conditional random fields to automatically add punctuation marks in the speech recognition output.
 - * The method obtained significant improvement, more than 10% absolute NIST error reduction, over the state-of-the-art method on English and Malay broadcast news.
- Collaborate with academic and industrial partners:
 - * Collaborate with researchers from Audio, Speech & Language Processing Group in NWPW China on speech segmentation tasks. Our work has been presented and published in 2 INTERSPEECH and APSIPA papers.
 - * Collaborate with researchers from A*STAR and DSO National Laboratories Singapore on a multi-lingual speech recognition system to collect online speech data, recognize the audio, and add punctuation to the ASR output.
- Administrative tasks to support the research team:
 - * Prepare seminars to transfer software systems to users.
 - * Plan and organize workshops and seminars with researchers from Microsoft, Google, and local universities on speech recognition and language processing topics.
 - * Organize annual team-building activities and events.
 - * Coordinate the publication of the conference proceedings, mobile applications, and online archive of INTERSPEECH 2014.

TEACHING
EXPERIENCE

• **Aalto University, Finland**

– Teaching Assistant.

Spring, 2018, 2019

- * Guest lecture
- * Design project work
- * Organize student discussion groups
- Supervise student X

• **Tampere University, Finland**

– Teaching Assistant.

Spring, 2018, 2019

- * Design coursework, home exercises, project works, and exam papers for SGN14007

Audio Processing and advanced level SGN24007.

* Guest lectures: Automatic Speech Recognition with Deep Neural Networks.

- Supervise Master student, Matti Tuhola, on thesis titled “Automatic Lexical Stress Recognition Using Neural Network”. This work invented a novel decoder to locate the stressed vowel in an English word and a sentence. This information assists English learners to pronounce words correctly. 2018
- Supervise Bachelor student, Marta Diaz Benito, on thesis titled “Language Modeling and Text Analysis on Lyrics in Popular Music”. This work developed a statistical model to classify the genre and artist of pop music lyrics in the past 20 years. 2017

• **Nanyang Technological University, Singapore**

- Advise Ph.D. student, Chenglin Xu, on a NLP project which focused on sentence segmentation. The project developed an automatic sentence unit detector to improve the readability of speech recognition output. 2014
- Advise Master student, Wenda Chen, in speech recognition project which focused on pronunciation modeling. The project developed an automatic scoring system to assist the user to learn standard English pronunciations. 2013
- Teaching Assistant Semester 1, 2011/2012
 - * Deliver tutorials for EE2006 Engineering Mathematics I and EE2076 Power Electronics.
 - * Obtained mean teaching score of 4.0/5.0 from the 4 tutorial and lab groups.
 - * Organize study groups to answer students’ requests for extra help on their coursework, besides the official teaching time of 63 hours.
- Supervise Bachelor students, Terrence Ho and Wye Deck Tan, on their final year projects. The project developed a voice command system using CMU Sphinx to control the motion of a robotic toy-cart. 2010

PROFESSIONAL
ACTIVITIES

- **Technical Committee** INTERSPEECH 2020, 2019.
- **Publication Committee** INTERSPEECH 2014.
- **Reviewer**
 - FCAI 2023
 - EURASIP Journal on Audio, Speech, and Music Processing
 - Speech Communication
 - Computer Speech and Language
 - ICASSP 2023
 - INTERSPEECH 2020, 2019, 2018, 2014, 2013
 - IEEE International Conference on Control, Automation, Robotics and Vision 2013, 2011

PROFESSIONAL
MEMBERSHIPS

- International Speech Communication Association (ISCA)
- Institute of Electrical and Electronics Engineers (IEEE)

REFERENCES

(Reference Letters Available Upon Request)

- **Prof. Mikko Kurimo**
Professor
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E-mail: mikko.kurimo@aalto.fi
- **Dr. Peter Sarlin**
CEO Silo.AI
Helsinki, Finland
E-mail: lpeter@siloi.ai
- **Prof. Tuomas Virtanen**
Research Fellow

Audio Research Group, Tampere University, Finland

E-mail: tuomas.virtanen@tuni.fi

- **Dr. Lori Lamel**

Principal Scientist

LIMSI-CNRS, France

E-mail: lamel@limsi.fr

- **Dr. Haizhou Li**

Director, Principal Scientist

Institute for Infocomm Research, A*STAR, Singapore

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- **Prof. Meng-Joo Er** (Ph.D. Thesis Advisor)

Professor

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